Phase 7 / Step 7.12: Molecular baseline (D2, NIST WebBook diatomic constants)

State (ground)	X 1Σg+ 1sσ2
T_e	0 cm^{-1}
ω_e	3115.500000 cm^{-1} (≈0.386273 eV)
ω_e x_e	61.820000 cm^{-1} (≈0.007665 eV)
B_e	30.4436000 cm^{-1} (≈0.00377453 eV)
α_e	1.0786000 cm^{-1} (≈0.00013373 eV)
D_e (distortion)	0.0114100 cm^{-1} (≈0.0000014147 eV)
r_e	0.741520 Å (=7.415200e-11 m)
Morse (derived)	D_e≈4.8667 eV, D0≈4.6755 eV (from ωe, ωexe)

Data: NIST Chemistry WebBook (Huber & Herzberg compilation)
This figure fixes baseline targets for Part III; it is not a P-model derivation.